

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Terminal Points and Interface Data - Power Island

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Rev	Date	Description
0	2026-09-27	Issued for EPC engineering (reference data)
A	2026-09-28	TQ-IEC-021: ST valve casing design, IF-45/IF-46 added, IF-03 location, IPB tap-off rating, agreed operating envelopes and dates

CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. Terminal points between IEC and the EPC

Rev A incorporates TQ-IEC-021: ST valve casing design conditions at IF-06/IF-10/IF-12, new terminal points IF-45/IF-46 (attenuation spray water), IF-03 located at the water injection skid inlet (upstream of the IEC injection pumps), IPB tap-off short-circuit rating (IF-36), and the operating envelopes and dates agreed with the EPC. Pressures in barg (IEC bar(a) values less 1.013 bar). The operating envelope is the range the providing side maintains at the point in all PER-003 cases; the EPC side is designed for at least the design conditions stated.

Tag	Service	IEC side	EPC side	Size / connection	Design p / T	Operating envelope at the point	Flow normal / max
IF-01	Natural gas to GT	performance gas heater inlet flange	gas from EPC gas receiving/metering station	12 in (DN300) CL600; ASME B16.5 CL600 RF weld-neck flange	63.0 barg / 80 degC	36.5 to 41.0 barg; 5-40 degC	20.67 / 23.00 kg/s
IF-02	LDO to GT	liquid fuel skid inlet flange	LDO forwarding/treatment system	6 in (DN150) CL150; ASME B16.5 CL150 RF flange	16.0 barg / 60 degC	3.0 to 6.0 barg; 10-45 degC	22.41 / 24.00 kg/s
IF-03	Demin water for water injection	water injection skid inlet	demin water storage/pumps	4 in (DN100) CL150; ASME B16.5 CL150 RF flange	16.0 barg / 50 degC	3.0 to 6.0 barg; 10-40 degC	13.44 / 15.00 kg/s
IF-04	Compressor wash water	wash skid inlet	demin water	2 in (DN50) CL150; ASME B16.5 CL150 RF flange	10.0 barg / 50 degC	3.0 to 6.0 barg; 10-40 degC	5.00 / 10.00 m3/h
IF-05	HP steam	HRSG HP SH outlet stop valve, weld end	main steam piping (EPC)	12 in P91 (SA-335 P91); butt-weld end ASME B16.25, IEC pup piece 300 mm	209.0 barg / 610 degC	59.0 to 192.0 barg; 560-600 degC	100.16 / 105.00 kg/s
IF-06	HP steam	ST main stop valve inlet, weld end	main steam piping (EPC)	12 in P91 (SA-335 P91); butt-weld end ASME B16.25, IEC pup piece 300 mm	214.0 barg / 610 degC	52.0 to 186.0 barg; 560-600 degC	100.16 / 105.00 kg/s
IF-07	Cold reheat	HP turbine exhaust, weld end	cold reheat piping (EPC)	24 in SA-335 P22; butt-weld end, IEC pup piece	51.0 barg / 420 degC	18.0 to 41.0 barg; 340-395 degC	100.16 / 105.00 kg/s
IF-08	Cold reheat	HRSG reheater inlet, weld end	cold reheat piping (EPC)	24 in SA-335 P22; butt-weld end, IEC pup piece	51.0 barg / 420 degC	17.0 to 41.0 barg; 340-395 degC	100.16 / 105.00 kg/s
IF-09	Hot reheat	HRSG reheater outlet stop valve, weld end	hot reheat piping (EPC)	28 in P91 (SA-335 P91); butt-weld end, IEC pup piece	49.0 barg / 610 degC	17.0 to 38.5 barg; 550-600 degC	117.50 / 125.00 kg/s
IF-10	Hot reheat	ST IP stop-control valve inlet	hot reheat piping (EPC)	28 in P91 (SA-335 P91); butt-weld end, IEC pup piece	51.0 barg / 610 degC	15.5 to 38.5 barg; 550-600 degC	117.50 / 125.00 kg/s
IF-11	LP steam	HRSG LP SH outlet stop valve	LP steam piping (EPC)	20 in SA-106 C; butt-weld end	9.0 barg / 330 degC	1.5 to 5.0 barg; 210-260 degC	11.39 / 16.50 kg/s
IF-12	LP steam	ST LP admission valve inlet	LP steam piping (EPC)	20 in SA-106 C; butt-weld end	9.0 barg / 330 degC	1.1 to 5.0 barg; 210-260 degC	11.39 / 16.50 kg/s
IF-13	HP / IP / LP bypass valves	valves supplied loose by IEC (incl. desuperheating)	installation in EPC piping, spray water from BFP/CEP (EPC)	HP 12 in / IP 28 in / LP 20 in (per valve datasheets); butt-weld ends; spray water flanges CL2500 (HP) / CL600 (IP, LP)	214.0 barg / 610 degC	1.0 to 192.0 barg; 150-600 degC	- / 125.00 kg/s
IF-14	Bypass steam to condenser	condenser neck dump devices (IEC)	IP/LP bypass downstream piping (EPC)	2 x 36 in + 1 x 28 in; butt-weld end at the condenser neck dump devices	15.0 barg / 250 degC	-1.0 to 12.0 barg; 40-250 degC	- / 141.50 kg/s
IF-15	Condensate from hotwell	hotwell outlet nozzle	CEP suction piping (EPC)	20 in (DN500) CL150; ASME B16.5 CL150 RF flange, vacuum-tight	2.5 barg / 80 degC	-0.995 to -0.90 barg; 15-40 degC	128.00 / 140.00 kg/s
IF-16	Condensate to CPH	HRSG CPH inlet	condensate line (EPC)	12 in (DN300) CL300; ASME B16.5 CL300 RF flange	24.0 barg / 180 degC	12.0 to 21.0 barg; 15-45 degC	141.37 / 150.00 kg/s
IF-17	BFP suction	HRSG LP drum downcomer outlet nozzle	BFP suction piping (EPC)	16 in (DN400) CL150; ASME B16.5 CL150 RF flange	9.0 barg / 185 degC	1.5 to 6.0 barg; 130-165 degC	129.98 / 140.00 kg/s
IF-18	HP feedwater	HRSG HP ECO1 inlet	HP feedwater piping (EPC)	10 in (DN250) CL2500; butt-weld end (SA-106 C / SA-335 P11)	234.0 barg / 200 degC	95.0 to 215.0 barg; 130-170 degC	100.16 / 110.00 kg/s
IF-19	IP feedwater	HRSG IP ECO inlet	IP feedwater piping (EPC)	6 in (DN150) CL600; butt-weld end	59.0 barg / 200 degC	20.0 to 48.0 barg; 130-165 degC	29.82 / 33.00 kg/s
IF-20	FGH water return	performance gas heater water outlet	return line to condensate (EPC)	4 in (DN100) CL600; ASME B16.5 CL600 RF flange	59.0 barg / 260 degC	18.0 to 46.0 barg; 60-90 degC	12.48 / 14.50 kg/s
IF-21	HRSG blowdown	continuous / intermittent blowdown valves outlet	blowdown tanks (EPC)	2 in / 3 in; socket-weld / butt-weld ends	209.0 barg / 370 degC	0.000 to 5.0 barg; 100-370 degC	1.00 / 5.00 kg/s
IF-22	HRSG drains and vents	drain/vent valves outlet (IEC)	drain collection, flash tanks, silencers (EPC)	various (1 in - 3 in); socket-weld / butt-weld ends	209.0 barg / 610 degC	0.000 to 1.0 barg; 40-610 degC	- / 10.00 kg/s
IF-23	Chemical dosing	dosing nozzles on drums/condensate	dosing skids and lines (EPC)	1 in; socket-weld, IEC root valve	234.0 barg / 100 degC	3.0 to 215.0 barg; 5-40 degC	- / 0.10 m3/h
IF-24	Sampling	sample nozzles with root valves (IEC)	sample lines, coolers, SWAS (EPC)	1/2 in; socket-weld, IEC root valve	234.0 barg / 610 degC	-1.0 to 215.0 barg; 20-600 degC	- / 0.50 kg/s
IF-25	Aqueous ammonia	ammonia injection skid inlet (IEC)	storage, unloading, forwarding pumps (EPC)	1 in; ASME B16.5 CL150 RF flange (316L)	10.0 barg / 40 degC	3.0 to 7.0 barg; 5-40 degC	0.12 / 0.50 t/h
IF-26	Flue gas to stack	HRSG outlet transition flange	stack (EPC), CEMS (EPC)	D 9.2 m; bolted flange with EPC fabric expansion joint	0.05 barg / 200 degC	0.000 to 0.03 barg; 60-120 degC	796.70 / 850.00 kg/s
IF-27	Circulating water	condenser water box flanges (2 inlets, 2 outlets)	CW pipes (EPC)	4 x DN1800; EN 1092-1 PN6 flanges, rubber-lined water boxes	4.0 barg / 40 degC	0.500 to 3.0 barg; 5-35 degC	36,050.00 / 38,000.00 m3/h
IF-28	Closed cooling water	IEC coolers inlet/outlet flanges	CCW system (EPC)	various (DN50 - DN400); ASME B16.5 CL150 RF flanges	10.0 barg / 60 degC	3.0 to 6.0 barg; 20-36 degC	1,055.00 / 1,160.00 m3/h
IF-29	Instrument and service air	IEC skids	compressed air system (EPC)	1 in - 2 in; ASME B16.5 CL150 RF flanges	10.0 barg / 50 degC	6.0 to 8.0 barg; 5-45 degC	350.00 / 500.00 Nm3/h
IF-30	Nitrogen	HRSG lay-up / generator purge connections	N2 supply (EPC)	1 in; ASME B16.5 CL150 RF flange	10.0 barg / 50 degC	5.0 to 8.0 barg; 5-40 degC	- / 1,000.00 Nm3/h
IF-31	Hydrogen / CO2	GT generator gas control panel inlet	bottle racks and piping (EPC)	1/2 in; compression fitting (316L)	16.0 barg / 50 degC	8.0 to 12.0 barg; 5-40 degC	- / 60.00 Nm3/h

Tag	Service	IEC side	EPC side	Size / connection	Design p / T	Operating envelope at the point	Flow normal / max
IF-32	Auxiliary steam	gland steam header inlet (IEC)	auxiliary boiler steam (EPC)	4 in (DN100) CL300; butt-weld end	15.0 barg / 320 degC	8.0 to 9.5 barg; 250-300 degC	4.50 / 7.50 t/h
IF-33	Fire protection	GT enclosure CO2 system (IEC)	fire detection/alarm, transformer deluge, hydrants (EPC)	-; hardwired volt-free contacts + Modbus TCP summary	-	-	-
IF-34	GT GSU HV	GT GSU HV bushings	380 kV connection to GIS (EPC)	560 MVA; GSU HV air bushings, EPC 420 kV connection to the GIS	-	380.00 kV (342.00-420.00 kV); Isc 50 kA, rating 50 kA	- / 560.00 MVA
IF-35	ST GSU HV	ST GSU HV bushings	380 kV connection to GIS (EPC)	270 MVA; GSU HV air bushings, EPC 420 kV connection to the GIS	-	380.00 kV (342.00-420.00 kV); Isc 50 kA, rating 50 kA	- / 270.00 MVA
IF-36	UAT supply	IPB tap-off flange (GT and ST IPB)	UAT and MV busduct (EPC)	21 kV (GT) / 15.75 kV (ST); IPB tap-off flange (IEC), EPC MV busduct	-	21.00 kV (19.95-22.05 kV); Isc 169 kA, rating 200 kA	- / 31.50 MVA
IF-37	MV supplies	SFC isolation transformer, IEC MV consumers	10.5 kV switchgear feeders (EPC)	10.5 kV feeders; cable terminations at the IEC equipment (lugs by EPC)	-	10.50 kV (9.45-11.55 kV); Isc 35 kA, rating 40 kA	- / 10.00 MVA
IF-38	LV supplies	IEC MCCs incoming	0.4 kV switchgear feeders (EPC)	0.4 kV feeders; IEC MCC incoming terminals	-	0.40 kV (0.36-0.44 kV); Isc 50 kA, rating 50 kA	-
IF-39	DC and UPS	TCS, protection, emergency oil pumps	220 V DC and 230 V AC UPS (EPC)	220 V DC / 230 V AC UPS; IEC panel incoming terminals	-	0.22 kV (0.19-0.24 kV)	-
IF-40	Control system interface	IEC turbine control system (TCS) gateway	plant DCS (EPC)	-; OPC UA + Modbus TCP, redundant fibre; hardwired trips	-	-	-
IF-41	Protection interface	generator/GSU protection panels	switchyard protection and TSO SCADA (EPC)	-; IEC 61850 (GOOSE/MMS) + hardwired trips	-	-	-
IF-42	Earthing	equipment earthing bosses	earthing grid (EPC)	-; bolted earthing pads (IEC), EPC 240 mm2 Cu risers	-	-	-
IF-43	Foundations	anchor bolts, sole plates, load data (IEC)	foundation design and construction (EPC)	-; anchor bolts / sole plates (IEC supply), grout by EPC to IEC specification	-	-	-
IF-44	Condenser tube cleaning balls	ball injection nozzles on the inlet water boxes	tube cleaning system (EPC)	2 x DN150 per inlet half; EN 1092-1 PN10 flanges	4.0 barg / 40 degC	0.500 to 3.5 barg; 5-35 degC	- / 200.00 m3/h
IF-45	HP attemperation spray water	HP attemperator spray control station	spray water line from the BFP discharge	3 in (DN80) CL2500; butt-weld end	234.0 barg / 200 degC	95.0 to 215.0 barg; 130-170 degC	- / 5.00 kg/s
IF-46	Reheat attemperation spray water	reheat attemperator spray control station	spray water line from the BFP interstage	2 in (DN50) CL600; butt-weld end	59.0 barg / 200 degC	20.0 to 48.0 barg; 130-170 degC	- / 3.00 kg/s

2. Scope break, isolation, protection and dates

Available = date the providing side has its side ready; needed = date the receiving side needs it (EPC commissioning sequence DEC-EPCE-0004).

Tag	Scope break	Isolation	Protection	Available / needed
IF-01	IEC supplies the performance gas heater inlet flange; EPC supplies the counter-flange, gasket (spiral wound, 316L/graphite), stud bolts and the pipe support nearest the flange, and makes the joint. Hydrotest up to the flange by each party; joint leak test with gas-in by EPC.	EPC: gas station outlet ESD valve (fail-closed, SIL 2, closed by the TCS gas trip) and manual valve with spectacle blind upstream of IF-01. IEC: gas skid stop and control valves.	EPC gas station pressure control: monitor + active regulator, slam-shut set 46 barg, token relief 44 barg; downstream design 63 barg (EN 12186 / ASME B31.8).	2028-08-31 / 2028-09-18
IF-02	IEC supplies the liquid fuel skid inlet flange; EPC supplies counter-flange, gasket, bolts, the last pipe support and the joint.	EPC: LDO ESD valve (fail-closed) and manual valve with spectacle blind at the skid boundary. IEC: skid stop valve.	EPC forwarding pumps: discharge relief to tank set 12 barg (< 16 barg design).	2028-11-15 / 2028-12-01
IF-03	IEC supplies the water injection skid inlet flange (skid with high-pressure pumps by IEC); EPC supplies counter-flange, gasket, bolts and the demin water forwarding pumps.	EPC manual isolation valve and spectacle blind; IEC skid inlet valve.	EPC demin forwarding pumps shut-off head <= 12 barg (< 16 barg design).	2028-11-15 / 2028-12-01
IF-04	IEC wash skid inlet flange; EPC demin water line, counter-flange, gasket and bolts.	EPC manual valve; IEC skid inlet valve.	EPC demin pumps shut-off <= 8 barg.	2028-09-15 / 2028-12-15
IF-05	IEC supplies the HP SH outlet stop valve with a 300 mm pup piece (matching material and wall); the field weld, its PWHT and 100 % RT/UT are by EPC, witnessed by IEC.	IEC HP SH outlet motor-operated stop valve and non-return valve.	IEC HRSG HP SH safety valves set <= 196 bar(a), electromatic relief 194 bar(a).	2028-09-30 / 2028-10-16
IF-06	IEC supplies the ST main stop valve with pup piece; field weld, PWHT and NDT by EPC.	IEC main stop valve (ST trip).	Upstream HRSG HP SH safety valves (IF-05) <= 196 bar(a); MSV casing design 215 bar(a) / 610 degC (TQ-IEC-021 item 1).	2028-09-30 / 2028-10-16
IF-07	IEC supplies the HP turbine exhaust nozzle with pup piece and the CRH non-return valve; field weld by EPC.	IEC CRH non-return valve (power assisted).	Reheater safety valves on the HRSG (IEC).	2028-09-30 / 2028-10-16
IF-08	IEC supplies the HRSG reheater inlet with pup piece and the CRH safety valves; field weld by EPC.	None at the HRSG inlet (CRH non-return valve at IF-07).	IEC cold reheat safety valves set 50 bar(a).	2028-09-30 / 2028-10-16
IF-09	IEC supplies the HRSG reheater outlet stop valve with pup piece; field weld, PWHT, NDT by EPC.	IEC HRH outlet stop valve.	IEC hot reheat safety valves set <= 48 bar(a).	2028-09-30 / 2028-10-16
IF-10	IEC supplies the ST IP stop/control valve with pup piece; field weld, PWHT, NDT by EPC.	IEC IP stop valve (ST trip).	Upstream HRH safety valves (IF-09) <= 48 bar(a); IP valve casing design 52 bar(a) / 610 degC (TQ-IEC-021 item 1).	2028-09-30 / 2028-10-16
IF-11	IEC supplies the LP SH outlet stop valve with pup piece; field weld by EPC.	IEC LP SH outlet stop valve and non-return valve.	IEC LP SH safety valve set 8.0 bar(a).	2028-09-30 / 2028-10-16
IF-12	IEC supplies the ST LP admission valve with pup piece; field weld by EPC.	IEC LP admission stop valve.	Upstream LP SH safety valve (IF-11) 8.0 bar(a); admission valve casing design 10 bar(a) / 330 degC (TQ-IEC-021 item 1).	2028-09-30 / 2028-10-16

Tag	Scope break	Isolation	Protection	Available / needed
IF-13	Loose supply: IEC supplies HP, IP and LP bypass valves with desuperheaters (free issue to site); EPC installs them in its piping, supplies the spray water lines from the BFP discharge/interstage and CEP discharge and all welds.	EPC isolation on the spray water lines; bypass valves close on loss of spray water (IEC logic).	HP bypass downstream design 52 bar(a) (CRH); IP/LP bypass downstream 16 bar(a) (IF-14).	2028-02-01 / 2028-03-01
IF-14	IEC supplies the dump devices in the condenser neck with weld ends; EPC supplies the IP/LP bypass downstream piping and the welds.	Bypass valves (IF-13).	Condenser rupture discs (IEC) and bypass trip on high condenser pressure.	2028-05-01 / 2028-06-01
IF-15	IEC hotwell outlet nozzle flange; EPC counter-flange, spiral-wound gasket, bolts and CEP suction piping.	EPC motor-operated suction valve per CEP.	Full vacuum design both sides.	2028-09-01 / 2028-09-15
IF-16	IEC HRSG condensate preheater inlet flange; EPC condensate line, counter-flange, gasket, bolts.	EPC motor-operated isolation valve.	EPC CEP shut-off head <= 22 bar(a) at maximum speed (< 25 bar(a) design), verified at pump FAT.	2028-09-01 / 2028-09-15
IF-17	IEC LP drum downcomer outlet nozzle flange; EPC BFP suction piping, counter-flange, gasket, bolts.	EPC suction isolation valve per BFP.	IEC LP drum safety valves.	2028-09-01 / 2028-09-15
IF-18	IEC HRSG HP economiser 1 inlet with pup piece; field weld by EPC.	EPC HP feedwater motor-operated isolation valve and non-return valve.	EPC BFP shut-off head at maximum speed <= 234 barg (VFD speed limit), verified at pump FAT; minimum-flow recirculation valve.	2028-09-01 / 2028-09-15
IF-19	IEC HRSG IP economiser inlet with pup piece; field weld by EPC.	EPC IP feedwater isolation and non-return valve.	EPC BFP interstage pressure at shut-off <= 58 bar(a), verified at FAT.	2028-09-01 / 2028-09-15
IF-20	IEC performance gas heater water outlet flange; EPC return line to the condensate system.	EPC isolation valve; IEC FGH outlet control valve.	EPC return line designed 60 bar(a) / 260 degC up to and including its pressure-reducing valve at the condensate connection; condensate system relief valve downstream (EPC).	2028-09-01 / 2028-09-15
IF-21	IEC supplies the continuous and intermittent blowdown valves (two in series); scope break at the outlet weld of the second valve; EPC blowdown piping, flash tanks and silencers.	IEC two valves in series (ASME I PG-58.3.6).	EPC blowdown piping designed for the drum design pressure up to the flash tank inlet.	2028-09-01 / 2028-09-15
IF-22	Scope break at the outlet weld of the IEC drain/vent valve (second isolation); EPC drain collection, atmospheric flash tanks and vent silencers.	IEC two valves in series on all pressure part drains.	EPC drain headers designed for the highest upstream design up to the first flash tank.	2028-09-01 / 2028-09-15
IF-23	IEC dosing nozzles with root valves on the drums; EPC dosing skids and lines.	IEC root valve + EPC isolation and check valve.	EPC dosing pump relief set below the line design.	2028-09-01 / 2028-09-15
IF-24	IEC sample nozzles with root valves; EPC sample lines, coolers and the SWAS.	IEC root valve; EPC isolation at the SWAS.	EPC sample line design = upstream design.	2028-09-01 / 2028-09-15
IF-25	IEC ammonia injection skid inlet flange; EPC storage, unloading and forwarding pumps.	EPC ESD valve (fail-closed) and manual valve.	EPC forwarding pump relief 9 barg.	2028-11-01 / 2028-11-15
IF-26	IEC HRSG outlet transition duct flange (drilled); EPC fabric expansion joint, stack, silencer, CEMS and damper (if any).	None (no stack damper).	HRSG casing design +/- 50 mbar.	2028-06-30 / 2028-09-15
IF-27	IEC condenser water box inlet/outlet flanges; EPC CW piping (GRP), expansion joints, counter-flanges and bolts.	EPC butterfly valves per inlet and outlet half (condenser half-side operation).	EPC CW pump shut-off head <= 3.5 barg at the condenser; surge analysis for pump trip <= 4 barg; vacuum breaker at the outlet high point.	2028-09-01 / 2028-09-15
IF-28	IEC cooler inlet/outlet flanges; EPC CCW system.	EPC valves per consumer.	EPC CCW pump shut-off <= 8 barg.	2028-08-09 / 2028-08-16
IF-29	IEC skid inlet flanges; EPC compressed air system.	EPC valve per skid.	EPC air receivers relief 10 barg.	2028-08-09 / 2028-08-16
IF-30	IEC lay-up / purge connections; EPC nitrogen supply.	EPC valve + IEC valve.	EPC pressure regulator + relief 9 barg.	2028-09-01 / 2028-09-15
IF-31	IEC generator gas control panel inlet; EPC bottle racks, regulators and piping.	EPC valves at the racks.	EPC regulator + relief 14 barg.	2028-08-25 / 2028-09-01
IF-32	IEC gland steam header inlet; EPC auxiliary steam line from the auxiliary boiler.	EPC isolation valve.	EPC auxiliary steam header relief set 15 barg.	2028-09-30 / 2028-10-16
IF-33	IEC GT enclosure CO2 fire protection panel terminals; EPC fire alarm panel (NFPA 72), cables, transformer deluge, hydrants.	-	-	2028-08-17 / 2028-08-24
IF-34	IEC GSU HV bushing terminals; EPC conductors/cable, clamps, surge arresters at the bushings and the GIS bay.	EPC GIS generator transformer bay circuit breaker, disconnect and earthing switch; IEC GCB on the LV side.	GSU differential (87T) and overall differential (87G-T) by IEC; GIS bay protection by EPC; trip matrix per IF-41.	2028-07-29 / 2028-08-02
IF-35	As IF-34 for the ST GSU.	As IF-34.	As IF-34.	2028-07-29 / 2028-08-02
IF-36	IEC IPB tap-off flange between GCB and GSU; EPC UAT, UAT busduct and flexible links.	IEC tap-off disconnect link; EPC UAT MV circuit breaker.	UAT differential and overcurrent by EPC, integrated in the IEC unit trip matrix.	2028-08-09 / 2028-08-16
IF-37	EPC 10.5 kV feeders and cables up to and including the terminations at the IEC equipment.	EPC feeder circuit breaker with earthing switch.	EPC feeder protection.	2028-08-09 / 2028-08-16
IF-38	EPC 0.4 kV feeders and cables up to the IEC MCC incoming terminals.	EPC feeder breaker; IEC MCC incomer.	EPC feeder protection selective with IEC MCC.	2028-08-09 / 2028-08-16
IF-39	EPC DC and UPS distribution feeders up to the IEC panel terminals.	EPC feeder MCB/fuse.	EPC selective fusing.	2028-08-09 / 2028-08-16
IF-40	IEC TCS gateway ports and fibre patch panel in the IEC electronic room; EPC fibre cables, DCS and the firewall between the zones (IEC 62443 zone/conduit).	-	-	2028-06-01 / 2028-06-15
IF-41	IEC generator/GSU protection panel terminals; EPC switchyard protection, substation automation and TSO SCADA.	-	-	2028-07-29 / 2028-08-02
IF-42	IEC equipment earthing pads and internal earthing; EPC earthing grid (IEEE 80) and risers.	-	-	2027-10-15 / 2027-11-15
IF-43	IEC load data, anchor bolts, sole plates and grouting specification; EPC foundation design, construction, cast-in of anchor bolts and grouting.	-	-	2027-11-01 / 2027-11-15

Tag	Scope break	Isolation	Protection	Available / needed
IF-44	IEC ball injection nozzles on the condenser inlet water boxes; EPC tube cleaning system (ball strainers in the outlet lines, recirculation pumps, collectors).	EPC valves at the nozzles.	As IF-27.	2028-09-01 / 2028-09-15
IF-45	IEC HP attemperator spray control station inlet (station with control and block valves by IEC); EPC spray water line from the BFP discharge and the weld.	EPC isolation valve at the BFP discharge header; IEC block valve.	As IF-18 (BFP shut-off <= 234 barg).	2028-09-01 / 2028-09-15
IF-46	IEC reheat attemperator spray control station inlet; EPC spray water line from the BFP interstage.	EPC isolation valve at the interstage header; IEC block valve.	As IF-19 (interstage <= 58 bar(a)).	2028-09-01 / 2028-09-15